



Monthly Intelligence Report

Colombia Edition 01 (Inaugural) — Bogotá D.C. & the Guavio Supply Corridor

Where Colombia operates a hydro-dominant Sistema Interconectado Nacional under XM dispatch — ~70% hydro in a normal year compressing to ~50% during a strong El Niño dry season — and where reservoir storage, not cooling-water demand, is the structural water-electricity coupling. The 1992–1993 Apagón (90-day rolling national rationing) anchors the cohort-worst hydro-drawdown event; the 2023–2024 strong El Niño + Q1 2024 Resolución CREG-101-009-2024 demand-curtailment regime mark the modern recurrence read; Nevado del Ruiz 1985 Armero (~23.000 fatalities) anchors a new R6_volcanic active reading second only in cohort to Iceland, and the durable R6_armed_conflict modifier — cohort-FIRST — folds in the structural ELN + dissident-FARC + Clan del Golfo pylon-attack residual along Catatumbo, Cauca-Nariño, and Bajo Cauca corridors. Inaugural edition · September 2026 · SSI v4.0.2

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SECTION A

Executive Summary — Fleet Overview

SUBSTATIONS SCORED

381

XM dispatch + ISA
INTERCOLOMBIA backbone
+ ~25 DSO operators · 33
departamentos (32 +
Bogotá D.C.)

GRID LINES MAPPED

~17.000 km

500 / 230 / 220 / 138 / 115 /
110 kV transmission
backbone

MC SIMULATIONS

3.81 M

10.000 per substation

PUBLIC DATA SOURCES

30+

95 variables · zero
proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). For Colombia, its engine processes ~42.000+ source data points per run, executes 3,81 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs ~465 million arithmetic operations — producing ~118.000+ output data points with full uncertainty quantification across 381 substations and ~17.000 km of 500 / 230 / 220 / 138 / 115 / 110 kV transmission spanning all 33 departamentos.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30+ verified public sources — XM Boletín Estadístico Mensual + ASIC market settlement data, CREG annual reliability + 2024 emergency demand-curtailment resolutions, UPME PERT 2024-2034 transmission expansion plan + PERG generation plan, DANE Cuentas Departamentales regional GDP + population, Banco de la República macro + COP/USD FX, IDEAM ENSO advisories + reservoir-level monitoring + flood-hazard maps, SGC seismic + 475-yr PGA hazard + per-volcano observatory bulletins (OVSM Manizales for Ruiz, OVSP Pasto for Galeras, OVSPop Popayán for Puracé), ColCERT + CSIRT-XM sectoral electricity CSIRT, Unidad de Víctimas IDP register + Migración Colombia refugee data, ANLA environmental licensing — making the methodology fully auditable and reproducible. Colombia is the inaugural single-country onboarding of Session 39, the first Andean OECD member in the SSI v4 cohort, the first onboarding to introduce the R6_armed_conflict durable modifier and the reservoir-electricity-nexus form of R6_drought, and the second cohort country to set R6_volcanic active (after Iceland, with a substantially higher catastrophic anchor in the 1985 Armero lahar event).

Substation in Focus

Subestación Cereté

Córdoba, cordoba · 110 kV

0.565

High

0.088

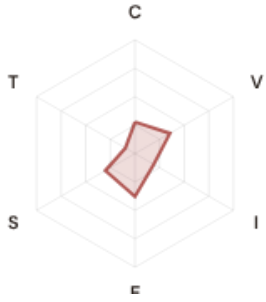
9th

R_FINAL

CLASSIFICATION

CI WIDTH

FLEET PERCENTILE



COMPONENTS

C Continuity: 0.278 / 0.30 (93%)

I Infrastructure: 0.191 / 0.25 (76%)

S Saturation: 0.301 / 0.20 (151%)

V Voltage: 0.355 / 0.10 (355%)

E Economic: 0.380 / 0.10 (380%)

T Transition: 0.097 / 0.05 (194%)

MODIFIERS

R3 Consequence: 1.027 ↑ amplifies

R4 Graph Criticality: 1.000 → neutral

R6a Restoration: 1.105 ↑ amplifies

R6b Seismic: 0.050 ↓ dampens

R7 Digital Readiness: 1.028 ↑ amplifies

This month's spotlight — automatically selected for June 2026 — is **Subestación Cereté** in Córdoba, cordoba. With an R_final of 0.565 (High band, 9th fleet percentile), its risk profile is dominated by the E (Economic) component at 380% of its weight ceiling, followed by V (Voltage) at 355%. Modifiers collectively shift R_base by 41.5%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor — SAIDI / SAIFI & the 1992-1993 Apagón Anchor

Why this section exists: Traditional grid risk frameworks treat reliability, cyber posture, and economic vulnerability as separate domains. The SSI's R7 (cyber posture) modifier, R6_drought (reservoir drawdown) α , and E (Economic) component cross those silos — revealing where structural electricity-supply stress intersects with consequence-of-loss. Colombia's reliability story is anchored by two recurring events: the 1992-1993 Apagón (90-day rolling national rationing triggered by El Niño-induced reservoir drawdown — Law 142 + Law 143 of 1994 unbundled the sector and created CREG + UPME + XM as a direct institutional response), and the 2023-2024 strong El Niño that drove reservoir levels below 35% by March 2024 and triggered the emergency Resolución CREG-101-009-2024 demand-response + IPP-curtailment + reservoir-conservation regime in Q1 2024. SAIDI varies sharply by departamento — Bogotá D.C. $\approx$ 5-8 hr/yr (urban Codensa cohort), Antioquia + Cundinamarca $\approx$ 10-20 hr/yr (regional capitals + EPM portfolio), Andean intermediate departamentos $\approx$ 20-50 hr/yr (Boyacá / Caldas / Tolima / Huila), Pacific Chocó + Amazon-frontier 50-150 hr/yr (cohort-worst rural reliability gap). ColCERT (MinTIC + MinDef joint, 2011) + the sectoral CSIRT-XM (operated by XM since 2019 under CONPES 3854/2016 mandate) + the Cyber Defence Command (CCOCI) under MinDef constitute the national cyber stack; no major sectoral cyber incident on the electrical grid is recorded as of mid-2024, placing R7 ceiling at 1,030 — mid-cohort, below the OECD-Americas crisis-hardened reference position.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

190

High/Critical + R7 < median

AVG R7 MODIFIER

1.0249

Range [0.99, 1.030]

HIGH-CONSEQUENCE SUBS

96

25.2% · R3 $\geq$ 1.05

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 cyber posture (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low cyber posture — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest. For Colombia the population is structurally moderate: CSIRT-XM coverage of the wholesale market and large generators is comprehensive since 2019, but the fragmented ~25-DSO distribution market means cyber

maturity drops sharply outside the Codensa + EPM + EPSA-Celsia + Air-e-Afinia mainline; the Chocó / Amazon-frontier cohort sits in the structural blind-spot quadrant.

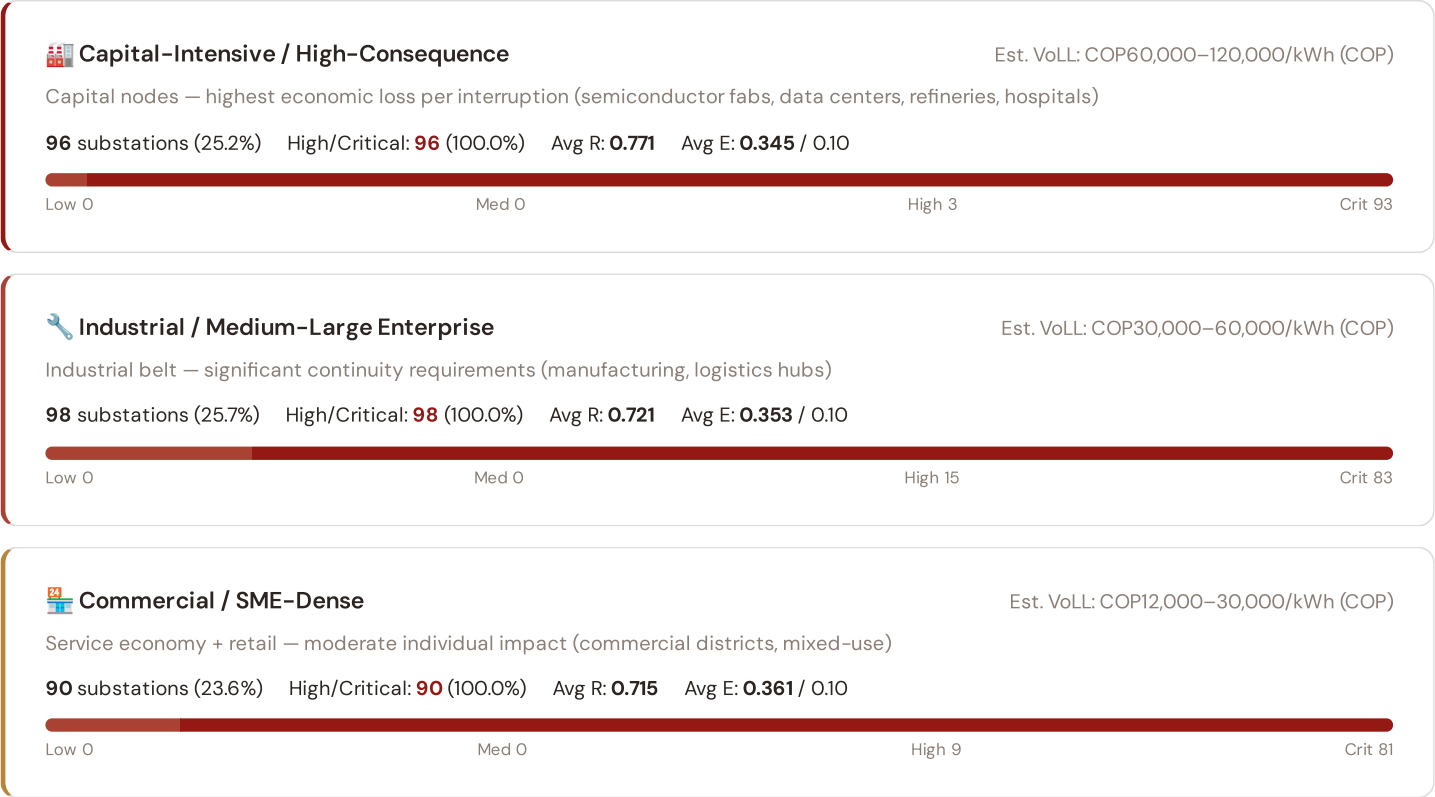


● Low ● Medium ● High ● Critical Dashed lines = fleet medians

The cyber-physical matrix identifies **190 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness ($R7 < 1.0248$). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **antioquia (29), bogota (26), valle-del-cauca (17)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Colombia's 5-tier R3 (Bogotá D.C. + Antioquia 1.05 / Regional-Capital 1.04 / Andean Intermediate 1.03 / Pacific-Frontier 1.02 / Amazon-Frontier Sparse 1.01) tracks the steep cordillera GDP gradient: Bogotá D.C. ≈\$11k GDP/cap nominal vs Chocó ≈\$2-3k, a ~5× spread that is cohort-leading in dispersion among recently-onboarded OECD members.



Sparse activity, agricultural land — lower VoLL (residential, agricultural)

97 substations (25.5%) High/Critical: 97 (100.0%) Avg R: 0.650 Avg E: 0.375 / 0.10



Value of Lost Load (VoLL) context: ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **96 substations** combine capital-intensive economic fabric ($R3 \geq 1.05$) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **antioquia (30), bogota (28), valle-del-cauca (11)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 12.6%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Departamento Cyber Posture Ranking

The 33 departamentos ranked by average R7 modifier — Colombia's gradient is driven by infrastructure concentration: Bogotá D.C. + Antioquia + Valle del Cauca + Santander host the bulk of CSIRT-XM-connected operations and the major DSO + generator NOCs, while the Pacific + Orinoquía + Amazonia cohort sits in a structurally-thinner cyber posture profile reflecting the lower digital-infrastructure density. Cross-referenced with average R_median, the Chocó / Vichada / Guainía / Vaupés / Amazonas fringe carries the largest combined grid-stress + cyber-thinness penalty.



WORST DIGITAL READINESS — TOP 15 DEPARTAMENTO REGIONS

Longer bar = higher R7 = better digital infrastructure.

1	Putumayo	1.0228
2	Cundinamarca	1.0237
3	Casanare	1.0240
4	Caldas ▲ high R	1.0244
5	Boyacá	1.0244
6	La Guajira	1.0245
7	Valle del Cauca ▲ high R	1.0246
8	Guaviare	1.0246
9	Cauca	1.0247
10	Antioquia ▲ high R	1.0247
11	Córdoba	1.0248
12	Norte de Santander	1.0249
13	Huila	1.0250
14	Santander ▲ high R	1.0251
15	Bogotá D.C. ▲ high R	1.0252

Departamento-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Putumayo** has the lowest average R7 (1.0228), while **Quindío** leads at 1.0266 — a spread of 0.0039 across the R7 range. **3 departamento regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (June 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0249, median = 1.0248; 0 substations classified HIGH cyber-exposure; 190 blind-spot substations (High/Critical risk + below-median R7); 162 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **13 verified public data sources** feeding 98 variables across 6 components and 9 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.



Data Source Registry — June 2026

d01_xm	XM — Compañía de Expertos en Mercados (ISA subsidiary; system operator CND + market operator ASIC since June 1995)	DAILY	C2,C3,I1,I2,I3,R4,T1	16 vars
d06_ideam	IDEAM — Instituto de Hidrología, Meteorología y Estudios Ambientales (hydro-meteorology + ENSO advisories)	DAILY	I2,I3,R6_climate,R6_...	8 vars
d03_sgc	SGC — Servicio Geológico Colombiano (seismic + volcanic monitoring)	CONTINUOUS	I1,I2,I3,R6_seismic,R...	14 vars
d05_osm	OSM Overpass — grid topology (ISO3166-1=CO area filter; 381 substations + line geometry per D#28)	MONTHLY	I1,I2,I3,R4	8 vars
d07_copernicus	Copernicus ERA5 + CMIP6 + UNGRD	MONTHLY	R6_climate	5 vars
d02b_dso	~25 DSOs (Codensa Enel + EPM municipal + EPSA-Celsia + Air-e/Afinia post-Electricaribe + CHEC/EBSA/ESSA/CENS/EDEQ/CEDENAR/DISPAC/ELECTROHUILA/EMCALI + ~10 more)	QUARTERLY	C1,V1,V2,I2,I3	18 vars
d04_dane	DANE — Departamento Administrativo Nacional de Estadística	QUARTERLY	E1,E2,S1,S2,S3	16 vars
d04b_banrep	Banco de la República — independent constitutional central bank (COP monetary authority)	QUARTERLY	E1,E2	5 vars
d08_uv_mc	Unidad de Víctimas (IDP register) + Migración Colombia (Venezuelan refugee data)	QUARTERLY	S1,S2,S3	6 vars
d01b_isa	ISA INTERCOLOMBIA — transmission asset owner (~70% of STN 220+ kV; BVC-listed ISA majority via Ministerio de Hacienda)	ANNUAL	I1,I2,I3,R4	8 vars
d02_creg	CREG — Comisión de Regulación de Energía y Gas (statutory regulator under MME; tariffs + reliability standards + market rules)	ANNUAL	C1,C2,V1,V2,T1,E1	12 vars
d09_colcert	CoLCERT + CSIRT-XM (sectoral electrical) + CCOCI (Cyber Defence Command, MinDef)	ANNUAL	R7_cyber	6 vars
d10_iaa_oecd	IEA + OECD — energy benchmarks (Colombia 37th OECD member, accession 28 Apr 2020)	ANNUAL	T1,E1	6 vars
d11_mme_upme	MME + UPME — Ministerio de Minas y Energía + Unidad de Planeación Minero-Energética	ANNUAL	T1,R6_drought	8 vars

C.1 Update Frequency Timeline



C.2 Fleet Score Snapshot



381

substations scored

0.733

P5=0.530 · P95=0.858

381

100.0% of fleet

100%

381 subs · CI/R < 0.50

BAND DISTRIBUTION



No primary data sources have been updated since the v4.0.2 launch baseline. All **381 substation scores remain unchanged** this edition. The fleet median R of 0.733 (mean 0.714) reflects a distribution skewed toward the lower bands, with 0.0% in Low and 0.0% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — Colombia Edition 01 (Session 39)

12 changes tracked across PO, P1, and P2 releases

C0-S39-1	NEW	Inaugural greenfield onboarding (Session 39, post-CR S33B + IL S35); first Andean OECD member; 37th OECD accession 28 Apr 2020	KB \$77
C0-S39-2	NEW	CO fact card \$1-\$19 web-verified anchors wired across all 8 pages (XM/ISA + CREG + MME/UPME + 33 departamentos + R6 modifiers)	KB \$77
C0-S39-3	NEW	R3 5-tier calibration — Capital-Metro/Industrial 1.05 (Bogotá D.C. + Antioquia) / Regional-Capital 1.04 (Cali + Bucaramanga + Cundinamarca + Atlántico + Bolívar + Casanare) / Andean Intermediate 1.03 (coffee zone + Galeras/Ruiz proximity) / Pacific-Frontier 1.02 (Chocó cohort-poorest + Cauca + La Guajira + Putumayo) / Amazon-Frontier Sparse 1.01 (Vichada/Guainía/Vaupés/Amazonas)	methodology
C0-S39-4	NEW	R6_drought NEW (reservoir-electricity nexus form) — first cohort hydro-reservoir-drawdown modifier (distinct from Israel desalination-electricity form). Anchors: 1992-1993 Apagón + 2015-2016 + 2023-2024 strong El Niño + CREG-101-009-2024 emergency	methodology
C0-S39-5	NEW	R6_armed_conflict NEW — first OECD cohort country with durable substation-proximate armed-conflict modifier. Heritage FARC-EP 1985-2002 + post-2016 ELN + dissidente FARC + Clan del Golfo (~40-80 pylon attacks/yr through 2024)	methodology
C0-S39-6	NEW	R6_volcanic cohort-second after Iceland S30 — Nevado del Ruiz Armero 1985 anchor (~23,000 fatalities, cohort-worst); 9 active edifices monitored by SGC (Ruiz/Galeras/Tolima/Puracé/Huila/Doña Juana/Cumbal/Sotará/Cerro Bravo); ~41 substations in proximity envelope	methodology
C0-S39-7	NEW	Ecuador 230 kV Pasto-Quito interconnect ACTIVE (~250 MW); Venezuela 230 kV Cuestecitas-Cuatricentenario SUSPENDED commercial flow since 2008; Panama Andean-CA interconnect extension Cerromatoso-Veladero planned	methodology
C0-S39-8	DATA	d05_osm LIVE — ISO3166-1=CO area filter (STN 500/230 kV ISA backbone + Ituango/Guavio/San Carlos/Sogamoso/El Quimbo switchyards) + D#28 `out geom` line-geometry compliance	methodology
C0-S39-9	DATA	XM TSO + ISA INTERCOLOMBIA + CREG regulator + ~25 fragmented DSO panel (Codensa Enel + EPM municipal + Air-e/Afinia + EPSA/Celsia + 20+ regional) + SGC + DANE + Banco de la República + IDEAM + UV + MC + CoLCERT/CSIRT-XM + UPME/MME wired	methodology
C0-S39-10	ENHANCED	edition_anchor_month_offset=5 — cohort-synchronized Edition 02 = 2027-02-11 (BPG Discipline #20 PASS)	intelligence
C0-S39-11	ENHANCED	D#21 content-leakage gate PASS — 0 CR proper-noun hits + 0 IL proper-noun hits (CO authored from COLOMBIA_FACT_CARD.md, A1c-at-content-layer fact-card-driven authoring per S33 Phase A lesson)	KB \$77
C0-S39-12	NEW	R7 ceiling 1.030 — baseline cyber-sectoral maturity (CONPES 3854/2016 + 3995/2020 + CoLCERT + CSIRT-XM + CCOCI); no major sectoral electrical cyber incident anchor	methodology

Bogotá D.C. — Capital District & the Guavio Supply Corridor

Deep-dive rotation: Each edition features a substantive thematic analysis. The departamento rotation covers Colombia's most material narratives: **Ed. 01** Bogotá D.C. — capital district + ~16% of national GDP + Guavio 1.200 MW reservoir critical supply for Cundinamarca (Edition 01 inaugural anchor) · **02** Antioquia — Medellín industrial heartland + EPM hydro portfolio + Ituango 2.400 MW (cohort-largest hydro project, 2024 full commissioning) · **03** Caldas — Nevado del Ruiz volcanic proximity + 1985 Armero anchor (~23.000 fatalities) + coffee-zone post-1999 Armenia earthquake recovery · **04** Atlántico — Barranquilla + Tebsa 914 MW (largest single thermal) + Caribbean dry-season backup during El Niño · **05** Chocó — Pacific rainforest (10.000 mm/yr wettest place on earth) + cohort-poorest social-resilience capacity + R6_armed_conflict residual along Bajo Cauca corridor.

D.1 Why Bogotá D.C., Why Now

CORRIDOR SUBSTATIONS

51

5 EHv · 46 HV

HIGH BAND

0 + 51

100.0% of corridor

CORRIDOR MEDIAN R

0.791

Fleet median: 0.733

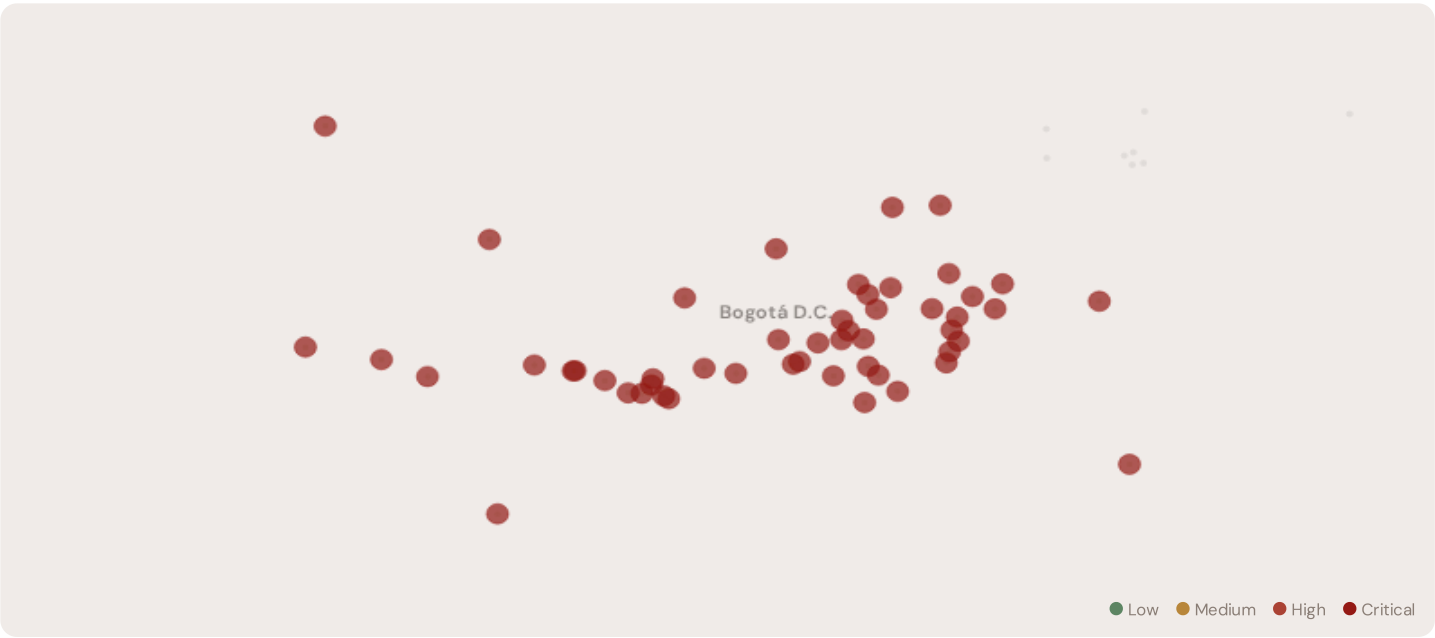
WORST DEPARTAMENTO

Bogotá D.C.

Avg R: 0.790 · 51 substations

The Bogotá D.C. — capital district + Guavio supply corridor (Edition 01 inaugural) hosts **51 substations** across 1 departamento regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **0 substations (0.0%)** are classified High and **51** Critical, with only **0** in the Low band. 100% of corridor substations score above the national fleet median of 0.733. The corridor median R of 0.791 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

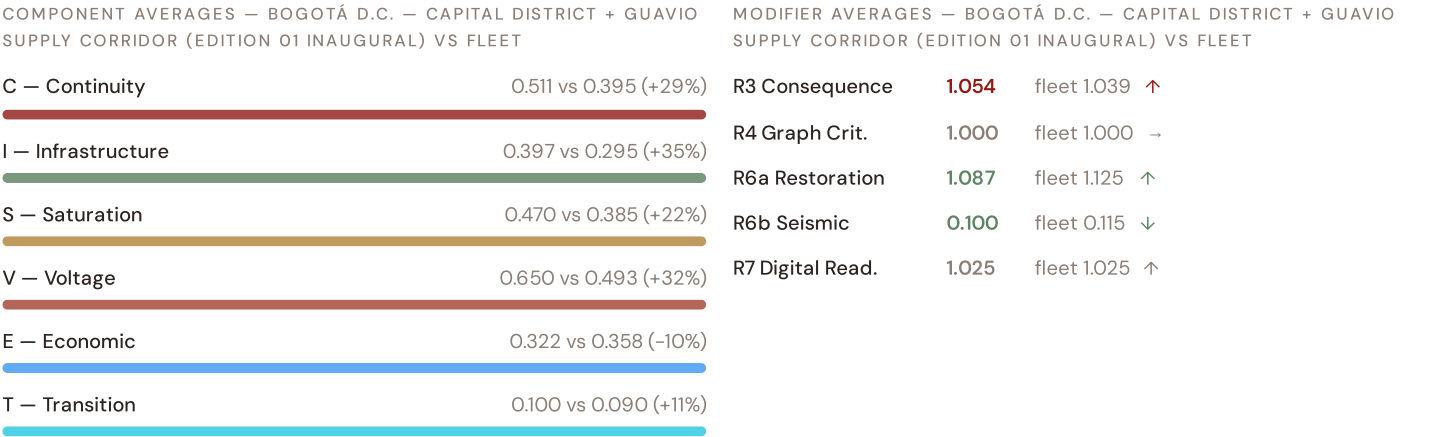
D.2 Corridor Map and Scoring



SUBSTATION	DEPARTAMENTO	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Subestación Compartir	Bogotá D.C.	0.848	Critical	0.533	0.697	0.430	0.347	0.501	0.107	
Salto I	Bogotá D.C.	0.841	Critical	0.551	0.643	0.445	0.349	0.498	0.102	
Subestación Bosa	Bogotá D.C.	0.840	Critical	0.514	0.685	0.414	0.354	0.492	0.114	
Concordia	Bogotá D.C.	0.838	Critical	0.527	0.699	0.365	0.353	0.486	0.115	
Calle 67	Bogotá D.C.	0.826	Critical	0.508	0.667	0.445	0.329	0.487	0.108	

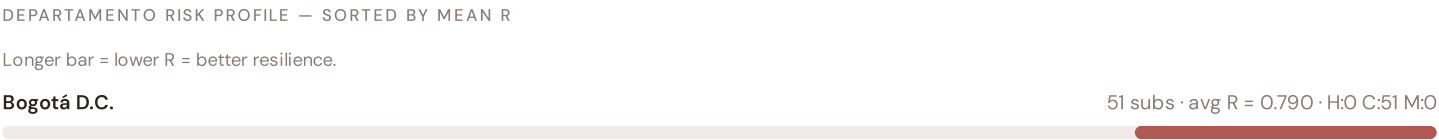
SUBSTATION	DEPARTAMENTO	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Carrera Quinta	Bogotá D.C.	0.826	Critical	0.544	0.692	0.385	0.337	0.490	0.104	
Subestación Fusagasuga – Sauces	Bogotá D.C.	0.824	Critical	0.465	0.721	0.394	0.344	0.481	0.110	
Subestación San Carlos	Bogotá D.C.	0.819	Critical	0.557	0.710	0.354	0.358	0.495	0.097	
Subestación El Río	Bogotá D.C.	0.815	Critical	0.554	0.678	0.399	0.331	0.490	0.103	
Florida–Bogotá	Bogotá D.C.	0.814	Critical	0.472	0.696	0.435	0.323	0.481	0.112	
... 41 additional substations — explore full corridor in Digital Twin —										

D.3 Component Analysis



The dominant risk driver across the Bogotá D.C. — capital district + Guavio supply corridor (Edition 01 inaugural) corridor is **V (Voltage)**, averaging 0.650 against a fleet average of 0.493 — occupying 650% of its weight ceiling (w = 0.10). The second contributor is **E (Economic)** at 0.322 vs fleet 0.358 (322% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.054 (fleet: 1.039), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 0.100, capturing the local seismic exposure — a dimension invisible to every competing framework.

D.4 Departamento Breakdown



The departamento regions-level breakdown reveals significant intra-regional variation. **Bogotá D.C.** leads with a mean R of 0.790 across 51 substations, while **Bogotá D.C.** scores 0.790 — a spread of 0.000 within a single corridor. The SSI's substation-level granularity reveals that the Bogotá D.C. — capital district + Guavio supply corridor (Edition 01 inaugural) is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

51 Bogotá D.C. — capital district + Guavio supply corridor (Edition 01 inaugural) substations score $R \geq 0.650$, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the Bogotá D.C. sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

OECD-Andean Context Card

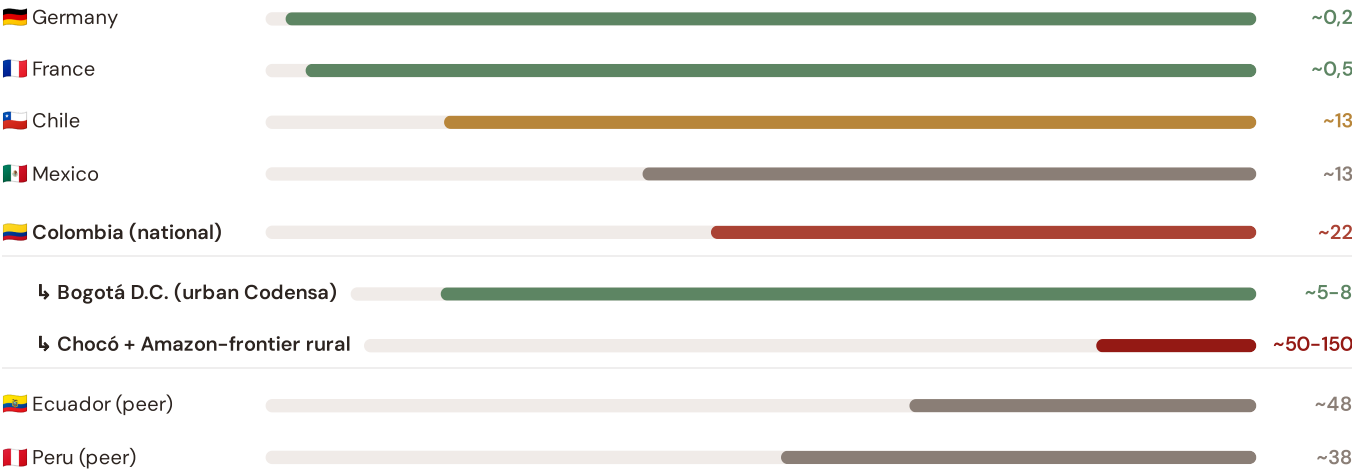
How this works: Each edition includes one peer-benchmarking card. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Bogotá D.C. + Guavio supply-corridor analysis hinges on Colombia's continuity performance relative to OECD-Americas peers and the broader Andean-region cohort. Future editions rotate through: VoLL comparison (Ed. 02), hydro-share + ENSO sensitivity vs Andean peers (03), CMIP6 drought hazard vs reservoir-electricity peers (04), etc. The underlying data updates annually with the CREG annual reliability report + the CIER Iberoamerican benchmarking cycle.

OECD-ANDEAN CONTEXT CARD — SAIDI BENCHMARKING

Ed. 01 · September 2026

Unplanned SAIDI (hours/customer/year) — Selected Countries

Longer bar = lower SAIDI = better reliability. Values in hours/customer/year (the Colombian cohort range demands hours; cohort-canonical minutes would overflow the bar visualisation).



Source: CREG annual reliability report (2024); CIER Iberoamerican benchmarking; per-DSO continuity filings (Codensa + EPM + EPSA-Celsia + Air-e + Afinia 2024). Departamento sub-categories from CREG territory classification (Bogotá D.C. urban Codensa vs Chocó + Amazon-frontier rural). Ecuador + Peru — Andean Community peer countries — included for regional benchmarking; Chile + Mexico as Latin-American OECD comparators. · Updated annually with CREG + CIER bulletin release.

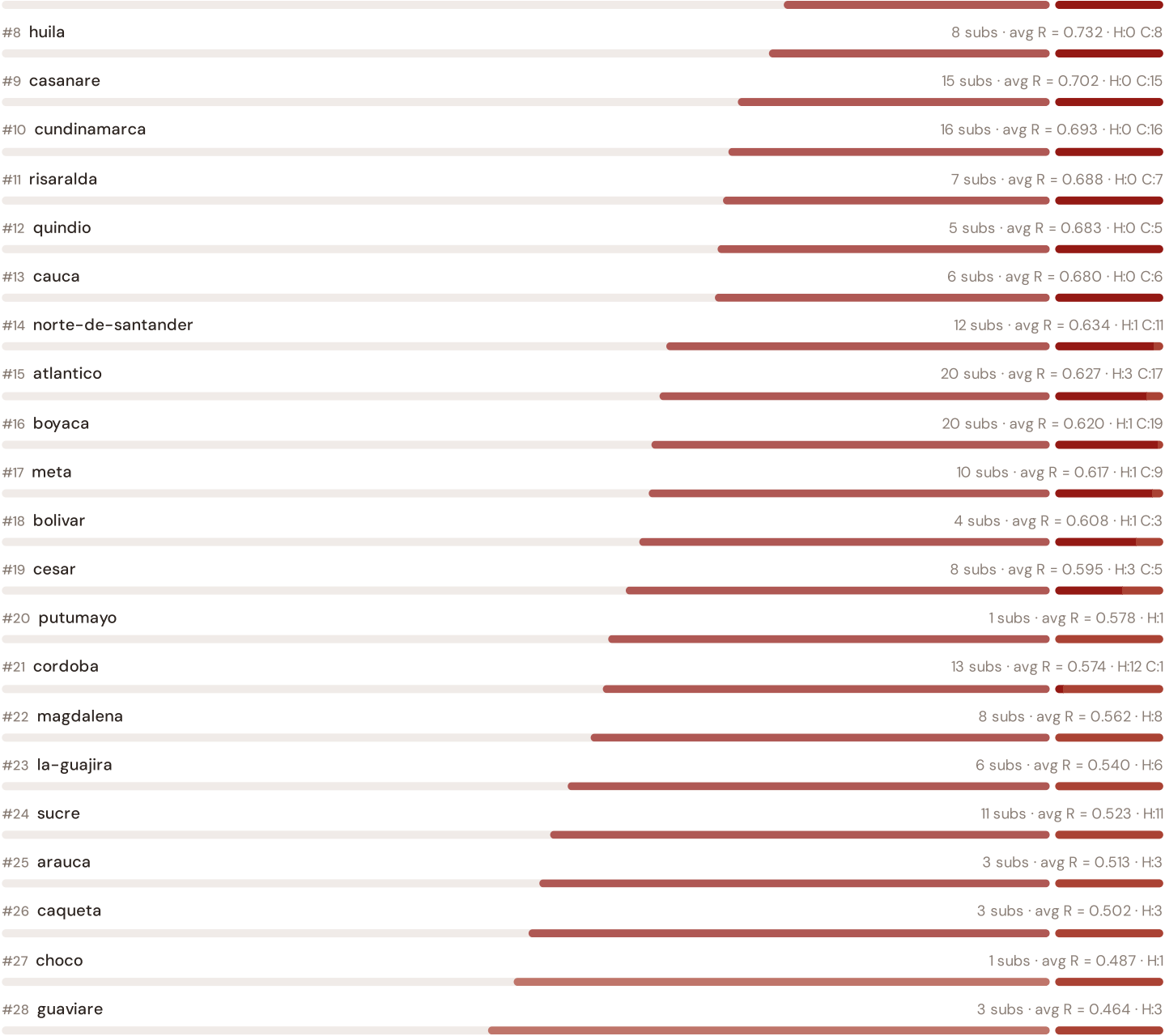
This month's key insight: The Bogotá D.C. — capital district + Guavio supply corridor (Edition 01 inaugural) corridor deep-dive shows **0 substations** (of 0) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.000 is **0.0x the fleet average** of 0.395. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Regional Risk Ranking — All 33 Colombian Departamentos

Where does Bogotá D.C. sit within the national landscape? Departamentos sorted by mean R_{median}, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



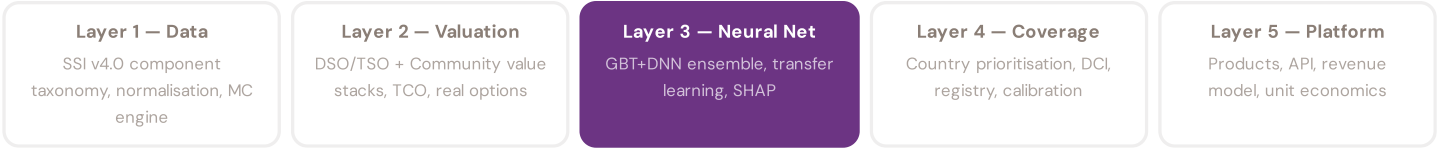


The Bogotá D.C. — capital district + Guavio supply corridor (Edition 01 inaugural) ranks **#0 of 28 departamento regions** by mean R_{median} , placing it among the upper tier of national risk. The three most stressed are antioquia, bogota, tolima, while the three most resilient are guaviare, choco, caqueta. 8 of 28 score above the fleet average of 0.714. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to XM as the system operator, to ISA INTERCOLOMBIA as the transmission asset owner, to EPM + Emgesa + Isagen + Celsia + AES Chivor as the generator cohort, to Codensa + EPM-distribution + EPSA-Celsia + Air-e + Afinia as the major DSOs, to CREG as the tariff regulator, and to the community?"* Each edition explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics — particularly relevant for Colombia given the 12 GW renewable-by-2030 UPME target requires significant battery storage to integrate the planned La Guajira wind cluster expansion + Cesar / Tolima / Atlántico solar build-out alongside the existing ~70%-hydro baseline.



Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history. Transparency converts model outputs into investment decisions.

KEY CONCEPTS

- SHAP: game-theoretic feature attribution per prediction
- Waterfall: fleet average → substation prediction
- Direct traceability to SSI components and data sources
- Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The training set comprises 381 substations with complete feature vectors. Average CI_width of 0.080 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 0 Low, 0 Medium, 58 High, 323 Critical — ensures balanced training across all risk bands.

Coming next month: **LAYER 3** Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

Looking Ahead

NEXT MONTH — EDITION 02

Deep-Dive: Atlántico

Deep-dive into Atlántico's grid risk profile — substation map, component analysis, departamento regions breakdown.

NEXT DATA REFRESH

Q3 2026 — 3 Quarterly Sources

DSO filings (Codensa + EPM + EPSA-Celsia + Air-e/Afinia + 20+ regional), DANE, Banco de la República + Unidad de Víctimas/Migración Colombia are due for refresh in Q3. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: CREG — Comisión de Regulación de Energía y Gas (statutory regulator under MME; tariffs + reliability standards + market rules) (12 variables → C1,C2,V1,V2,T1,E1).

FLEET SPOTLIGHT

Córdoba: Highest Risk Concentration

Córdoba has the highest concentration of High/Critical substations: **13 of 13 (100%)** are in the upper risk bands — avg R = 0.574. This makes it the most acute risk corridor in the fleet.

Edition 02 will be published on the second Thursday of July 2026. Deep-dive region: **Atlántico**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent XM, ISA INTERCOLOMBIA, CREG, MME, UPME, DANE, Banco de la República, IDEAM, SGC, UNGRD, ANLA, MinTIC, CoICERT, CSIRT-XM, EPM, Emgesa, Isagen, Celsia, AES Chivor, Codensa, EPSA, Air-e, Afinia, CHEC, EBSA, ESSA, CENS, EDEQ, CEDENAR, DISPAC, ELECTROHUILA, EMCALI, the Universidad Nacional de Colombia, the Universidad de los Andes, EAFIT, or any Colombian academic or industry institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Colombia Edition 01 (Inaugural)

Published 10 September 2026 · SSI Index v4.0.2 · 381 substations · ~17.000 km grid lines · 6 components · 20 metrics · 8 modifiers (incl. R6_armed_conflict NEW + R6_drought reservoir-electricity-nexus form + R6_volcanic active)
10.000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196.608 evaluations per substation) · CMIP6 SSP2-4.5